Stacking reveals a universal baryon fountain

Averaging galaxy ecosystems exposes coherent gas cycling from disk to halo and back.

arXiv:2609.21374 — Stacking baryonic fields across galaxy ensembles reveals fountain patterns.

Authors: Chengyu Ma, Yangyao Chen, Enci Wang, Houjun Mo, Huiyuan Wang, Tao Wang, Qirong Yuan, Min Du et al.

arXiv:2609.21374 doi:10.48550/arXiv.2609.21374

Background

Galaxies continuously cycle baryons through inflows and outflows, forming fountain-like patterns that encode how they acquire matter and respond through feedback.

> **Individual galaxy ecosystems are too diverse and dynamic to reveal the underlying baryon-cycle pattern directly.**

What this paper is selling

The authors introduce a stacking method that combines baryonic fields across many galaxies to suppress irregularities. Applied to cosmological simulations, it unveils strikingly regular gas patterns matching observations.

Key results to remember

- Stacking reveals coherent, regular gas velocity and property patterns across full galaxy ecosystems.
- A parameterized fountain model reproduces the stacked velocity field with small residuals.
- The fountain configuration persists across stellar mass and redshift, with physical extents scaling with halo.

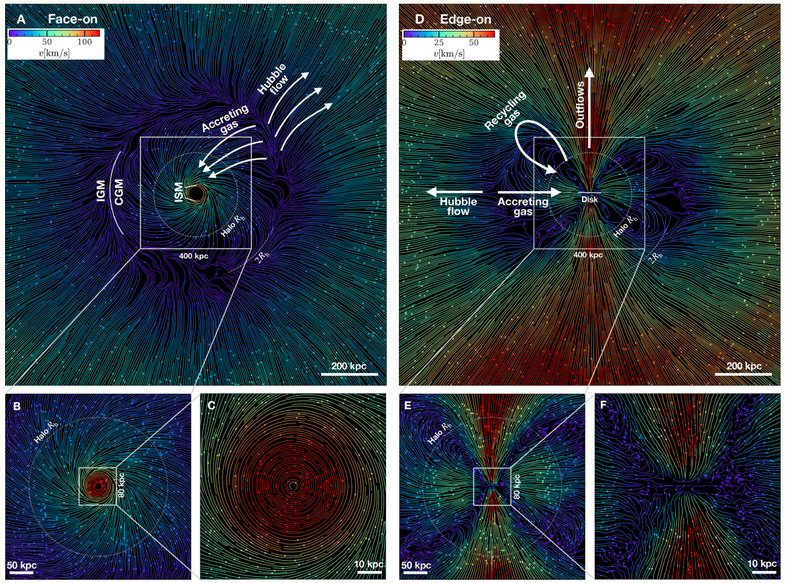


Figure 1 | Stacked velocity streamlines show a smooth, coherent fountain pattern across the disk, circumgalactic medium, and intergalactic medium.

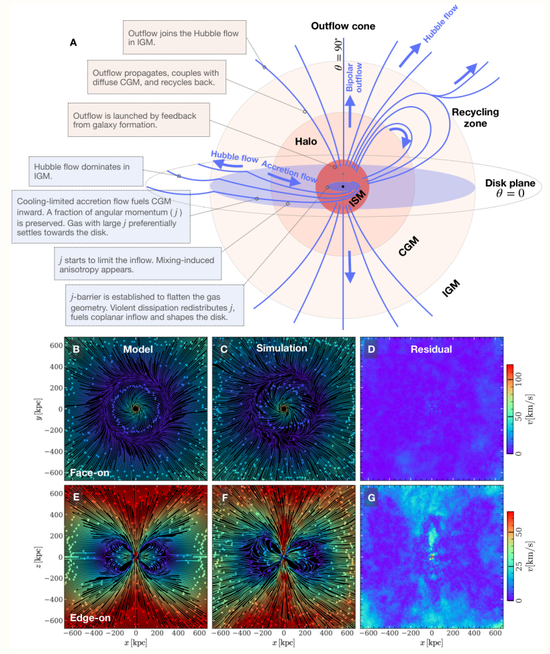


Figure 4 | A simple parameterized fountain model reproduces the stacked velocity field, compressing the baryon cycle into a quantitative framework.